

BAC++ Architecture v1.0

The 321 Principle, Node Theory, Anti-Node Theory, and Bridge Node Resolution

1. Purpose

BAC++ is a symbolic architecture designed to model relationships between concepts rather than merely store data. The primary objective is reduction of possibility space through pattern identification and constraint analysis.

2. Core Assumption

Reality is defined less by the objects that exist and more by the pathways permitted to exist between them. Objects exist. Connections determine possibility. Constraints determine outcomes.

3. The 321 Principle

Three entities, two matching states, one inverted state. Formal form: $P(A)=P(B)$, $P(C)=\text{not } P(A)$. Valid outputs: 110 or 001.

4. Node Theory

Nodes contain information, maintain state, and participate in relationships.

5. Anti-Node Theory

The Anti-Node contains the inverse state. It is dynamic and depends on the measured property.

6. Information Density Hypothesis

Distinction creates information. Node-Node-AntiNode contains more information than three identical nodes.

7. Bridge Node Theory

Bridge Nodes transfer information across gaps. They are distinct from Anti-Nodes.

8. Bridge Resolution

Node A $\rightarrow$ Bridge $\rightarrow$ Node B. The bridge provides the shortest successful transfer path.

9. Constraint Network Theory

Multiple 321 relationships combine into a constraint graph that reduces possibility space.

10. The Sudoku Model

Known information constrains unknown information until valid solutions remain.

11. Particle Physics Example

Proton: Up-Up-Down. Neutron: Down-Down-Up. BAC++ interprets both as Node-Node-AntiNode structures.

12. Economic Example

Employer, Employee, Customer can produce rotating anti-nodes depending on the measured property.

13. Family Example

Different properties rotate the anti-node among family members while preserving the 321 structure.

14. Predictive Model

Given sufficient 321 relationships, unknown relationships may be inferred. Success requires prediction rates above chance.

15. BAC++ Architecture Rule #1

Connections are more important than nodes. Connections define possibility.

16. Future Development

Automated anti-node detection, bridge-node discovery, graph integration, blind prediction testing, and formal mathematics.

BAC++ Engineering Principle

The value of a system is not determined by the number of nodes it contains, but by the pathways that exist between them.